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Detection device

Abstract

A detection device includes a substrate having a detection region, a coupling region, and a wiring region between the detection region and the coupling region, a plurality of detection elements disposed in the detection region of the substrate, a plurality of terminals disposed in the coupling region of the substrate and electrically coupled to the detection elements, a plurality of coupling wires adjacently disposed with a gap interposed therebetween in the wiring region of the substrate and electrically coupling the detection elements to the terminals, an insulating layer provided to the wiring region of the substrate and covering the coupling wires, and a plurality of through holes formed in the wiring region of the substrate and passing through the substrate and the insulating layer in a thickness direction between the coupling wires.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims the benefit of priority from Japanese Patent Application No. 2022-164571 filed on Oct. 13, 2022, the entire contents of which are incorporated herein by reference.
BACKGROUND
1. Technical Field

(2) The present invention relates to a detection device.
2. Description of the Related Art
(3) Japanese Patent Application Laid-open Publication No. 2019-029514 (JP-A-2019-029514) describes a wiring device including a rigid substrate, a stretchable substrate stacked on the rigid substrate, and stretchable wiring provided on the stretchable substrate. The wiring device described in JP-A-2019-029514 is used in strain sensors. Japanese Patent Application Laid-open Publication No. 2021-103298 (JP-A-2021-103298) describes a stretchable display device that can display images when warped or stretched. U.S. Unexamined Patent Application Publication No. 2016/0174304 (US 2016/0174304 A1) describes an OLED display device with a bendable region.
(4) Japanese Patent Application Laid-open Publication No. 2018-146489 (JP-A-2018-146489) describes a force sensor including a plurality of electrodes spaced apart from each other.
(5) In JP-A-2019-029514, JP-A-2021-103298, and US 2016/0174304 A1, a deformable part (e.g., stretchable substrate) and a non-deformable part (e.g., rigid substrate) are coupled. Stress may possibly be generated due to the state of deformation of the deformable part, thereby breaking the wiring. Alternatively, stress may possibly increase due to the state of coupling between the deformable part and the non-deformable part, thereby breaking the wiring.
(6) There is a demand for a detection device, such as the force sensor described in JP-A-2018-146489, to be attached to a member having a curved surface and detect the physical quantity, such as force. In this case, stress may possibly act on coupling wires extending from a plurality of force sensors, thereby breaking the wiring.
(7) An object of the present invention is to provide a detection device that can suppress breaking of a plurality of wires electrically coupled to a plurality of detection elements.
SUMMARY
(8) A detection device according to an embodiment of the present disclosure includes a substrate having a detection region, a coupling region, and a wiring region between the detection region and the coupling region, a plurality of detection elements disposed in the detection region of the substrate, a plurality of terminals disposed in the coupling region of the substrate and electrically coupled to the detection elements, a plurality of coupling wires adjacently disposed with a gap interposed therebetween in the wiring region of the substrate and electrically coupling the detection elements to the terminals, an insulating layer provided to the wiring region of the substrate and covering the coupling wires, and a plurality of through holes formed in the wiring region of the substrate and passing through the substrate and the insulating layer in a thickness direction between the coupling wires.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a plan view schematically illustrating a detection device according to a first embodiment;
(2) FIG. **2** is a view for explaining an example of the use of the detection device according to the first embodiment;
(3) FIG. **3** is a sectional view along line III-III′ of FIG. **2**;
(4) FIG. **4** is a sectional view along line IV-IV′ of FIG. **2**;
(5) FIG. **5** is a circuit diagram of the circuit configuration of force sensors according to the first embodiment;
(6) FIG. **6** is a sectional view along line VI-VI′ of FIG. **2**;
(7) FIG. **7** is a sectional view of the force sensor according to the first embodiment before force is input;
(8) FIG. **8** is a sectional view of the force sensor according to the first embodiment after force is

input;
(9) FIG. **9** is an enlarged plan view of a wiring region of the detection device according to the first embodiment;
(10) FIG. **10** is a sectional view along line X-X′ of FIG. **9**;
(11) FIG. **11** is a sectional view along line XI-XI′ of FIG. **9**;
(12) FIG. **12** is an enlarged plan view of a wiring region according to a first modification;
(13) FIG. **13** is an enlarged plan view of a wiring region according to a second modification;
(14) FIG. **14** is an enlarged plan view of a wiring region according to a third modification;
(15) FIG. **15** is an enlarged plan view of a wiring region according to a fourth modification;
(16) FIG. **16** is an enlarged plan view of a wiring region according to a fifth modification;
(17) FIG. **17** is an enlarged plan view of a wiring region according to a sixth modification;
(18) FIG. **18** is an enlarged plan view of a wiring region according to a seventh modification;
(19) FIG. **19** is an enlarged plan view of a wiring region according to an eighth modification;
(20) FIG. **20** is an enlarged plan view of a coupling wire in a wiring region according to a ninth modification;
(21) FIG. **21** is an enlarged plan view of a coupling wire in a wiring region according to a tenth modification;
(22) FIG. **22** is an enlarged plan view of a coupling wire in a wiring region according to an eleventh modification;
(23) FIG. **23** is an enlarged plan view of a wiring region according to a twelfth modification;
(24) FIG. **24** is a sectional view of a force sensor according to a thirteenth modification including parallel electrodes;
(25) FIG. **25** is a sectional view of a force sensor according to a fourteenth modification including hybrid electrodes;
(26) FIG. **26** is a sectional view of a force sensor according to a fifteenth modification before force is input;
(27) FIG. **27** is a sectional view of the force sensor according to the fifteenth modification after force is input;
(28) FIG. **28** is a sectional view of a force sensor according to a sixteenth modification before force is input;
(29) FIG. **29** is a sectional view of the force sensor according to the sixteenth modification after force is input;
(30) FIG. **30** is a sectional view of a detection device according to a seventeenth modification;
(31) FIG. **31** is a plan view of a detection device according to an eighteenth modification;
(32) FIG. **32** is a plan view of a detection device according to a nineteenth modification;
(33) FIG. **33** is a plan view of an example of the use of the detection device according to the nineteenth modification;
(34) FIG. **34** is a sectional view along line XXXIV-XXXIV′ of FIG. **33**; and
(35) FIG. **35** is a sectional view of a detection device according to a second embodiment.

DETAILED DESCRIPTION

(36) Exemplary aspects (embodiments) to embody the present invention are described below in greater detail with reference to the accompanying drawings. The contents described in the embodiments below are not intended to limit the present disclosure. Components described below include components easily conceivable by those skilled in the art and components substantially identical therewith. Furthermore, the components described below may be appropriately combined. What is disclosed herein is given by way of example only, and appropriate modifications made without departing from the spirit of the present disclosure and easily conceivable by those skilled in the art naturally fall within the scope of the present disclosure. To simplify the explanation, the drawings may possibly illustrate the width, the thickness, the shape, and other elements of each unit more schematically than those in the actual aspect. These elements, however, are given by way

of example only and are not intended to limit interpretation of the present disclosure. In the present disclosure and the drawings, components similar to those previously described with reference to previous drawings are denoted by like reference numerals, and detailed explanation thereof may be appropriately omitted.
(37) To describe an aspect regarding a certain structure on which another structure is disposed in the present specification and the claims, when “on” is simply used, it indicates both the following cases unless otherwise noted: a case where the other structure is disposed directly on and in contact with the certain structure, and a case where the other structure is disposed on the certain structure with yet another structure interposed therebetween.
First Embodiment
(38) FIG. **1** is a plan view schematically illustrating a detection device according to a first embodiment. A detection device **1** according to the first embodiment is a force sensor that detects force acting on an input surface **1***a* (refer to FIG. **6**). As illustrated in FIG. **1**, the detection device **1** includes an array substrate **2** (substrate **21**), a sensor unit **10**, a gate line drive circuit **15**, a signal line selection circuit **16**, a wiring substrate **110**, a control substrate **101**, and a drive integrated circuit (IC) **102**.
(39) The control substrate **101** is electrically coupled to the substrate **21** via the wiring substrate **110**. The wiring substrate **110** is a flexible printed circuit board or a rigid board, for example. The control substrate **101** is provided with the drive IC **102**. The drive IC **102** supplies control signals to the sensor unit **10**, the gate line drive circuit **15**, and the signal line selection circuit **16** to control the detection operation of the sensor unit **10**. The drive IC **102** may include a drive circuit that supplies voltage signals, such as power signals, to the sensor unit and the gate line drive circuit **15**. The drive IC **102** may include a detection circuit that receives detection signals output from a plurality of force sensors **3** and performs signal processing on the detection signals.
(40) The substrate **21** has a detection region AA, a peripheral region GA, a wiring region WA, and a coupling region CA. The detection region AA is a region provided with the plurality of force sensors **3**. The peripheral region GA is a region between the outer periphery of the detection region AA and the outer periphery of the substrate **21** and is not provided with the force sensors **3**. The gate line drive circuit **15** and the signal line selection circuit **16** are provided in the peripheral region GA of the substrate **21**. In the example illustrated in FIG. **1**, the peripheral region GA has a frame shape surrounding the detection region AA.
(41) In the following description, a first direction Dx is a direction in a plane parallel to the substrate **21**. A second direction Dy is a direction in the plane parallel to the substrate **21** and is orthogonal to the first direction Dx. The second direction Dy may intersect the first direction Dx without being orthogonal thereto. A third direction Dz is a direction orthogonal to the first direction Dx and the second direction Dy. The third direction Dz is the normal direction of the substrate **21**. The term “plan view” refers to the positional relation when viewed from a direction perpendicular to the substrate **21**.
(42) The force sensors **3** (detection elements) are arrayed in a matrix (row-column configuration) in the detection region AA. In other words, the force sensors **3** are arrayed in the first direction Dx and the second direction Dy in the detection region AA. The force sensors **3** of the sensor unit **10** are devices that detect force input to a plurality of individual detection regions obtained by dividing the detection region AA. The force sensors **3** output detection signals Vdet corresponding to the force input to them. The force sensors **3** perform detection due to gate drive signals supplied from the gate line drive circuit **15**. The force sensors **3** output electrical signals corresponding to the force input to them as detection signals Vdet to the signal line selection circuit **16**. The configuration of the force sensor **3** will be described later in greater detail with reference to FIG. **6**.
(43) The gate line drive circuit **15** is a circuit that drives a plurality of gate lines GL (refer to FIG. **5**) based on control signals supplied from the drive IC **102**. The gate line drive circuit **15** sequentially or simultaneously selects a plurality of gate lines GL and supplies gate drive signals to

the selected gate lines GL. Thus, the gate line drive circuit **15** selects the plurality of force sensors **3** coupled to the gate lines GL.

(44) The signal line selection circuit **16** is a switch circuit that sequentially or simultaneously selects a plurality of signal lines SL (refer to FIG. **5**). The signal line selection circuit **16** is a multiplexer, for example. The signal line selection circuit **16** selects the signal lines SL based on selection signals supplied from the drive IC **102**. As a result, the signal line selection circuit **16** electrically couples the force sensors **3** coupled to the selected signal lines SL to a coupling wire **17**. Thus, the detection signals Vdet from the force sensors **3** are output to the drive IC **102** via the coupling wire **17** and the wiring substrate **110**.

(45) The detection region AA, the peripheral region GA, the wiring region WA, and the coupling region CA are adjacently positioned in the second direction Dy. The coupling region CA of the substrate **21** is provided with a plurality of terminals T electrically coupled to the force sensors **3** via the signal line selection circuit **16** and the coupling wire **17**. The coupling region CA of the substrate **21** is coupled to the wiring substrate **110** via the terminals T. The width of the coupling region CA in the first direction Dx is smaller than that of the detection region AA and the peripheral region GA in the first direction Dx.

(46) The wiring region WA is positioned between the detection region AA and the coupling region CA in the second direction Dy. The wiring region WA is provided with a plurality of coupling wires **17** that electrically couple the force sensors **3** to the terminals T. In FIG. **1**, the boundary between the wiring region WA, and the detection region AA and the peripheral region GA is virtually represented by an alternate long and short dash line. The boundary between the wiring region WA and the coupling region CA is virtually represented by an alternate long and short dash line. In the example illustrated in FIG. **1**, the wiring region WA has a trapezoidal shape with a long side on the detection region AA side and the peripheral region GA side and a short side on the coupling region CA side. The configuration of the wiring region WA will be described later in greater detail with reference to FIG. **9**.

(47) FIG. **2** is a view for explaining an example of the use of the detection device according to the first embodiment. FIG. **3** is a sectional view along line III-III′ of FIG. **2**. FIG. **4** is a sectional view along line IV-IV′ of FIG. **2**. As illustrated in FIGS. **2** to **4**, the detection device **1** according to the first embodiment is used by being wrapped around the outer peripheral surface of a housing **120**. The housing **120** has a solid columnar shape with a curved outer peripheral surface. The housing **120**, however, does not necessarily have a columnar shape and may at least partially have a curved surface. The housing **120** is a handrail of stairs or a grip of sports equipment or a directional training machine, for example. The detection device **1** detects the magnitude and the distribution of force applied when a user grips the housing **120**.

(48) More specifically, as illustrated in FIG. **3**, the detection region AA and the peripheral region GA of the substrate **21** are provided to the curved surface of the housing **120** and are attached in a curved state along the outer peripheral surface. The substrate **21** is attached to the housing **120** by adhesive or double-sided tape, which is not illustrated. The detection device **1** is disposed with the second direction Dy in FIG. **1** along the extending direction (axial direction) of the housing **120** and with the first direction Dx in FIG. **1** along the circumferential direction of the housing **120**. The third direction Dz (normal direction of the substrate **21**) in FIG. **1** of the detection device **1** corresponds to the radial direction of the housing **120**.

(49) As illustrated in FIG. **4**, the outer peripheral surface of the housing **120** is provided with a base **121**. The coupling region CA of the substrate **21** is attached on the flat surface of the base **121** of the housing **120**. With this configuration, the coupling region CA is provided to the outer peripheral surface of the housing **120** in a flat state, thereby securing satisfactory coupling with the wiring substrate **110**.

(50) As illustrated in FIG. **2**, the wiring region WA of the substrate **21** is deformed to couple the detection region AA and the peripheral region GA in a curved state to the coupling region CA in a

flat state. The side of the wiring region WA on the detection region AA side is curved along the outer peripheral surface of the housing **120**. The side of the wiring region WA on the coupling region CA side is flat along the flat surface of the base **121**.
(51) The housing **120** illustrated in FIGS. **2** to **4** is given by way of example only, and the housing **120** may have a conical shape, for example. The detection device **1** is not necessarily attached to the outer peripheral surface of the housing **120** and may be attached to housings with other shapes. For example, the detection device **1** is not necessarily attached to a convex curved surface and may be attached to a concave curved surface. While FIGS. **1** to **4** illustrate an example where the wiring substrate **110** is coupled to the coupling region CA of the substrate **21**, the present embodiment is not limited thereto. Alternatively, the drive IC **102** may be mounted on the coupling region CA of the substrate **21**.
(52) The following describes an example of the circuit configuration and the operation of the detection device **1**. FIG. **5** is a circuit diagram of the circuit configuration of the force sensors according to the first embodiment. As illustrated in FIG. **5**, the gate lines GL each extend in the first direction Dx and are arrayed in the second direction Dy. The signal lines SL each extend in the second direction Dy and are arrayed in the first direction Dx. The detection device **1** also includes common wiring, which is not specifically illustrated, extending along the peripheral region GA. The common wiring is supplied with a reference potential Vref from the drive IC **102**.
(53) A drive transistor Tr is provided corresponding to each of the force sensors **3**. The drive transistor Tr is composed of a thin-film transistor (TFT) and is an re-channel metal oxide semiconductor (MOS) TFT in this example.
(54) The gate of the drive transistor Tr is coupled to the gate line GL. The source (source electrode **52** (refer to FIG. **6**)) of the drive transistor Tr is coupled to the force sensor **3**. The drain (drain electrode **53** (refer to FIG. **6**)) of the drive transistor Tr is coupled to the signal line SL.
(55) With this configuration, when the gate line drive circuit **15** scans the gate line GL, the detection signals Vdet corresponding to the force input to the force sensors **3** are obtained via the signal lines SL. As a result, the values of force acting on the individual detection regions corresponding to the respective force sensors **3** are obtained based on the magnitude of the detection signals Vdet obtained via the signal lines SL.
(56) The drive transistor Tr is not limited to an n-type TFT and may be a p-type TFT. A plurality of transistors may be provided corresponding to one force sensor **3**.
(57) The following describes the multilayered configuration of the detection device **1**. FIG. **6** is a sectional view along line VI-VI′ of FIG. **2**. In the following description, a force direction A**1** may be referred to as “the lower side” or simply as “under”, and the direction opposite to the force direction A**1** may be referred to as “the upper side” or simply as “on”.
(58) As illustrated in FIG. **6**, the detection region AA, the peripheral region GA, the wiring region WA, and the coupling region AA include the substrate **21** and a circuit layer **55** extending continuously and shared by them. The substrate **21** is an insulating and flexible substrate. The substrate **21** is made of a resin substrate or a resin film, for example. The circuit layer **55** is composed of circuit elements, such as the drive transistor Tr, various kinds of wiring coupled to the circuit elements, gate insulating films **22** and **22**A, and an insulating layer **23**.
(59) In the detection region AA, the drive transistor Tr, the insulating layer **23**, the force sensor **3**, and a protective layer **33** are stacked on the substrate **21**. In the peripheral region GA, a transistor Tr-S and the insulating layer **23** are provided on the substrate **21**. In other words, the force sensor **3** is not formed on the insulating layer **23** in the peripheral region GA. The transistor Tr-S in the peripheral region GA is a transistor that constitutes a peripheral circuit, such as the signal line selection circuit **16**, and is a switching element that switches coupling and decoupling between the signal lines SL and the coupling wire **17**, for example.
(60) In the wiring region WA and the coupling region CA, the coupling wire **17** and the insulating layer **23** are provided on the substrate **21**. The substrate **21** and the insulating layer **23** in the wiring

region WA and the coupling region CA are formed continuously with and made of the same material as that of the substrate **21** and the insulating layer **23** in the detection region AA and the peripheral region GA.

(61) The coupling wire **17** is continuously provided from the peripheral region GA to the wiring region WA and the coupling region CA. The coupling wire **17** electrically couples the force sensors **3** (detection elements) to the terminals T. Specifically, one end of the coupling wire **17** is coupled to the signal line selection circuit **16** (transistor Tr-S) in the peripheral region GA and is electrically coupled to the force sensor **3** via the signal line selection circuit **16** (transistor Tr-S) and the signal line SL. In the coupling region CA, a contact hole CH is formed in the region of the insulating layer **23** overlapping the coupling wire **17**. The terminal T in the coupling region CA is formed by the coupling wire **17** exposed at the bottom of the contact hole CH.

(62) In the detection region AA, the drive transistor Tr includes a semiconductor layer **51**, a source electrode **52**, a drain electrode **53**, and a gate electrode **54**. The semiconductor layer **51** is provided on the substrate **21**. The gate electrode **54** is provided on the semiconductor layer **51** with the gate insulating film **22** interposed therebetween. The source electrode **52** is coupled to the source region of the semiconductor layer **51**. The source electrode **52** is electrically coupled to a detection electrode **31** of the force sensor **3** through a contact hole formed in the insulating layer **23**. The drain electrode **53** is coupled to the drain region of the semiconductor layer **51**. The drain electrode **53** is electrically coupled to the signal line SL (refer to FIG. **5**).

(63) Similarly to the drive transistor Tr, the transistor Tr-S in the peripheral region GA includes a semiconductor layer **51**A, a source electrode **52**A, a drain electrode **53**A, and a gate electrode **54**A. The gate electrode **54**A is provided on the semiconductor layer **51**A with the gate insulating film **22**A interposed therebetween. The multilayered structure of the transistor Tr-S is the same as that of the drive transistor Tr described above, and repeated explanation thereof is omitted.

(64) The insulating layer **23** is provided on the substrate **21** to cover the drive transistor Tr. The insulating layer **23** covers the drive transistor Tr in the detection region AA and is continuously provided to the peripheral region GA, the wiring region WA, and the coupling region AA as described above. While FIG. **6** illustrates only two insulating layers of the gate insulating film **22** and the insulating layer **23** as the insulating layers of the circuit layer **55** to simplify the explanation, three or more insulating layers may be stacked.

(65) The force sensor **3** is provided on the insulating layer **23** of the circuit layer **55**. The force sensor **3** includes a sensor layer **30**, the detection electrode **31**, and a common electrode **32**. The detection electrode **31** is provided on the insulating layer **23**. The detection electrode **31** is made of conductive metal material. Alternatively, the detection electrode **31** may be made of translucent conductive material, such as indium tin oxide (ITO). The detection electrodes **31** are separated from each other corresponding to the respective force sensors **3**. In other words, the detection electrodes **31** are arrayed in a matrix (row-column configuration) in the detection region AA (refer to FIG. **1**).

(66) The common electrode **32** is disposed facing the detection electrodes **31**. The common electrode **32** is a solid film formed on the lower surface of the protective layer **33**. In other words, the common electrode **32** overlaps the entire detection region AA and is shared by the force sensors **3**. The common electrode **32** is made of conductive metal material. Alternatively, the common electrode **32** may be made of translucent conductive material, such as ITO. The common electrode **32** is coupled to common wiring (not illustrated) by wiring, which is not illustrated, and is supplied with the reference potential Vref from the drive IC **102**.

(67) The sensor layer **30** is provided between the detection electrode **31** and the common electrode **32**. The sensor layer according to the present embodiment is provided on the lower surface (surface facing the detection electrode **31**) of the common electrode **32**. The sensor layer **30** overlaps the entire detection region AA. The sensor layer **30** is made of conductive resin. The sensor layer **30** has a plurality of protrusions **30***a* on the lower surface (surface facing the detection electrode **31**).

Each protrusion **30***a* is separated from the detection electrode **31** and the insulating layer **23** with a space interposed therebetween when no force is applied to the input surface **1***a*.

(68) The protective layer **33** is provided covering the common electrode **32**. The upper surface of the protective layer **33** serves as the input surface **1***a* of the detection device **1**. Similarly to the substrate **21**, the protective layer **33** is an insulating and flexible substrate. The protective layer **33** is made of a resin substrate or a resin film, for example.

(69) FIG. **7** is a sectional view of the force sensor according to the first embodiment before force is input. FIG. **8** is a sectional view of the force sensor according to the first embodiment after force is input. FIGS. **7** and **8** do not illustrate the circuit layer **55**. As illustrated in FIG. **7**, some of the protrusions **30***a* are in contact with the detection electrode **31** before force is input. However, the number of protrusions **30***a* in contact with the detection electrode **31** is small, and the detection electrode **31** and the common electrode **32** facing across the sensor layer **30** remain insulated. FIG. **7** is given by way of example only, and the protrusions **30***a* may be separated from and not in contact with the detection electrode **31** before force is input.

(70) When the input surface **1***a* of the force sensor **3** is pressed, the protective layer **33** and the common electrode **32** deform and come closer to the detection electrode **31**. The sensor layer **30** moves in the force direction A**1**, and the protrusions **30***a* of the sensor layer **30** come into contact with the detection electrode **31**. As a result, the detection electrode **31** and the common electrode **32** are electrically coupled via the sensor layer **30**, and an electric current flows to the detection electrode **31**.

(71) As the force acting on the sensor layer **30** increases, the number of protrusions **30***a* in contact with the detection electrode **31** increases, and the contact area between the sensor layer **30** and the detection electrode **31** increases. In addition, the protrusions **30***a* are pressed against and planarized on the detection electrode **31**, thereby increasing the contact area with the detection electrode **31**. For this reason, the contact resistance between the sensor layer **30** and the detection electrode **31** decreases corresponding to an increase in force (increase in contact area), and the amount of electric current flowing to the detection electrode **31** increases. Therefore, the force sensor **3** can detect the force value based on the magnitude of the current value.

(72) The following describes the configuration of the wiring region WA in greater detail. FIG. **9** is an enlarged plan view of the wiring region of the detection device according to the first embodiment. FIG. **10** is a sectional view along line X-X′ of FIG. **9**. FIG. **11** is a sectional view along line XI-XI′ of FIG. **9**.

(73) As illustrated in FIG. **9**, the wiring region WA of the substrate **21** has a trapezoidal shape that couples the peripheral region GA (and the detection region AA) to the coupling region CA. In other words, the sides of the wiring region WA at the outer ends in the first direction Dx are inclined with respect to the second direction Dy. The width in the first direction Dx of the side of the wiring region WA on the coupling region CA side is smaller than the width in the first direction Dx of the side of the wiring region WA on the peripheral region GA side (and the detection region AA side).

(74) The coupling wires **17** in the wiring region WA are each linearly provided and are arrayed in the first direction Dx with a gap interposed therebetween. The coupling wires **17** extend along the second direction Dy at the center of the wiring region WA in the first direction Dx. The coupling wires **17** have a larger inclination angle with respect to the second direction Dy as closer to the outer ends of the wiring region WA in the first direction Dx. In other words, the arrangement pitch of the coupling wires **17** varies depending on the position in the second direction Dy. The arrangement pitch of the coupling wires **17** on the coupling region CA side is smaller than that of the coupling wires **17** on the peripheral region GA side (and the detection region AA side). The arrangement pitch of the coupling wires **17** arrayed in the first direction Dx is substantially constant at a predetermined position in the second direction Dy.

(75) In the wiring region WA of the substrate **21**, a plurality of through holes TH are formed between the coupling wires **17**. The through holes TH are each have a circular shape. The through

holes TH are formed corresponding to the respective coupling wires **17** in the first direction Dx. The through holes TH arrayed in the first direction Dx have the same diameter. The through holes TH are arrayed along the extending direction of the coupling wires **17**. The through holes TH have a larger opening area (diameter) as the distance between the coupling wires **17** adjacently disposed in the first direction Dx is larger. In other words, the opening area (diameter) of the through holes TH is larger on the peripheral region GA side (and the detection region AA side) and is smaller on the coupling region CA side along the extending direction of the coupling wires **17**.

(76) The coupling wires **17** and the through holes TH are formed to be line-symmetric with respect to an axis of symmetry AX that passes through the center of the wiring region WA in the first direction Dx and is parallel to the second direction Dy.

(77) As illustrated in FIG. **10**, the insulating layer **23** is provided covering the coupling wires **17**. The through holes TH pass through the substrate **21** and the insulating layer **23**. In other words, the through holes TH open on the upper surface of the insulating layer **23** and on the lower surface of the substrate **21**. As illustrated in FIG. **11**, the substrate **21** and the insulating layer **23** are continuously provided between the coupling wires **17** at the part where the through holes TH are not formed.

(78) As described above with reference to FIGS. **2** to **4**, the detection region AA, the peripheral region GA, and the wiring region WA of the substrate **21** are provided to the curved surface of the housing **120** with the first direction Dx along the circumferential direction, and the coupling region CA is provided to the flat surface of the base **121**. In this case, in the wiring region WA, compressive stress acts on the lower surface of the substrate **21** facing the outer peripheral surface of the housing **120** along the circumferential direction, and tensile stress acts on the upper surface of the insulating layer **23** on the opposite side. These stresses are relatively small at the center of the wiring region WA in the first direction Dx and become relatively larger toward the outer ends in the first direction Dx. This causes a stress distribution in the wiring region WA in the direction in which the coupling wires **17** are arrayed (first direction Dx). The wiring region WA has a part that deforms along the outer peripheral surface of the housing **120** and a part flatly provided on the coupling region CA side. This causes a stress distribution in the wiring region WA also in the extending direction of the coupling wires **17** (second direction Dx).

(79) The present embodiment can relieve the stress generated in the substrate **21** and the insulating layer **23** because the wiring region WA has a plurality of through holes TH. The substrate **21** and the insulating layer **23**, for example, have flexibility in deformation at the part where the through holes TH are formed. When stress is generated in the substrate **21** and the insulating layer **23**, the substrate **21** and the insulating layer **23** deform to reduce the stress. Therefore, the present embodiment can suppress the stress generated in the substrate **21** and the insulating layer **23** in the wiring region WA compared with the case where the through holes TH are not formed. As a result, the present embodiment can suppress breaking of the coupling wires **17** in the wiring region WA.

(80) As described above, the through holes TH are arrayed in the direction in which the coupling wires **17** are arrayed (first direction Dx), thereby suppressing the stress in the wiring region WA in the direction in which the coupling wires **17** are arrayed (first direction Dx). In addition, the through holes TH are arrayed along the extending direction of the coupling wires **17**, thereby also effectively suppressing the stress generated in the extending direction of the coupling wires **17**.

Modifications

(81) The shape, number, arrangement, and others of the through holes TH illustrated in FIG. **9** are given by way of example only and can be appropriately modified. The shape, number, arrangement, and others of the coupling wire **17** are given by way of example only and can be appropriately modified. In the following description, the same components as those described in the embodiment above are denoted by like reference numerals, and duplicate explanation thereof is omitted.

(82) FIG. **12** is an enlarged plan view of a wiring region according to a first modification. FIG. **13** is an enlarged plan view of a wiring region according to a second modification. As illustrated in

FIG. **12**, a through hole THa according to the first modification has a rectangular shape. Alternatively, as illustrated in FIG. **13**, a through hole THb according to the second modification has an elliptical shape. As described above, the shape of the through holes TH, THa, and THb can be appropriately modified. The through holes TH do not necessarily have the shapes in the examples according to the first and the second modifications and may have polygonal or other shapes. Alternatively, the through holes TH, THa, and THb having different shapes may be combined and formed in one wiring region WA.

(83) FIG. **14** is an enlarged plan view of a wiring region according to a third modification. As illustrated in FIG. **14**, a plurality of through holes THc according to the third modification are arrayed along the extending direction of the coupling wires **17** and have the same shape and the same opening area (diameter). A larger number of through holes THc are formed as the distance between the coupling wires **17** adjacently disposed in the first direction Dx is larger. More precisely, the number of through holes THc per unit area is larger on the peripheral region GA side (and the detection region AA side) and is smaller on the coupling region CA side along the extending direction of the coupling wires **17**.

(84) In FIG. **14**, four through holes THc are collectively formed between the adjacent coupling wires **17** on the peripheral region GA side (and the detection region AA side). Two through holes THc are collectively formed between the adjacent coupling wires **17** at the center in the extending direction of the coupling wires **17**. One through hole THc is formed between the adjacent coupling wires **17** on the coupling region CA side. The number of through holes THc illustrated in FIG. **14** is given by way of example only and can be appropriately modified.

(85) FIG. **15** is an enlarged plan view of a wiring region according to a fourth modification. As illustrated in FIG. **15**, a through hole THd according to the fourth modification has a slit shape. The through hole THd extends along the extending direction of the coupling wires **17**. In the fourth modification, one through hole THd is formed between the adjacent coupling wires **17**. The number, shape, arrangement, and others of the slit-like through holes THd can be appropriately modified.

(86) FIG. **16** is an enlarged plan view of a wiring region according to a fifth modification. As illustrated in FIG. **16**, a plurality of through holes THe according to the fifth modification are arrayed along the extending direction of the coupling wires **17**. The through holes THe are separated by a substrate **21***a* provided with no slit along the extending direction of the coupling wires **17**. In FIG. **16**, three through holes THe are formed between the coupling wires **17**. The present modification is not limited thereto, and the number of through holes THe arrayed along the extending direction of the coupling wires **17** can be appropriately modified. The number of through holes THe may vary depending on the position in the first direction Dx.

(87) FIG. **17** is an enlarged plan view of a wiring region according to a sixth modification. As illustrated in FIG. **17**, two through holes THf according to the sixth modification are formed side by side between the adjacent coupling wires **17**. While two through holes THf are formed between the coupling wires **17** in FIG. **17**, the number of through holes THf may be three or more. Alternatively, the number of through holes THf may vary depending on the position in the first direction Dx. For example, one through hole THf may be formed between the adjacent coupling wires **17** at the center of the wiring region WA in the first direction Dx, and the number of through holes THf may increase as closer to the outer ends of the wiring region WA in the first direction Dx.

(88) FIG. **18** is an enlarged plan view of a wiring region according to a seventh modification. As illustrated in FIG. **18**, a through hole THg according to the seventh modification has a first slit portion THga and second slit portions THgb. The first slit portion THga extends along the extending direction of the coupling wires **17**. The second slit portions THgb intersect the first slit portion THga. Three second slit portions THgb are formed corresponding to one first slit portion THga. The present modification is not limited thereto, and the number of second slit portions THgb

formed corresponding to one first slit portion THga can be appropriately modified.
(89) FIG. **19** is an enlarged plan view of a wiring region according to an eighth modification. As illustrated in FIG. **19**, a through hole THh according to the eighth modification are formed to remove the substrate **21** and the insulating layer **23** in most regions between the adjacent coupling wires **17**. In other words, the opening width of the through hole THh in the first direction Dx is substantially equal to the distance between the coupling wires **17** adjacently disposed in the first direction Dx. In one coupling wire **17**, the width in the direction intersecting the extending direction of the coupling wire **17** is substantially equal to the width of the substrate **21** to which the coupling wire **17** is provided. The present modification can suppress the stress generated in the substrate **21** and increase the flexibility in deformation of each of the coupling wires **17** provided to the substrate **21**.
(90) FIG. **20** is an enlarged plan view of a coupling wire in a wiring region according to a ninth modification. As illustrated in FIG. **20**, a coupling wire **17***a* according to the ninth modification has a zigzag line shape. More specifically, the coupling wire **17***a* has a plurality of linear portions **18***a*, and the linear portions **18***a* are coupled by bends **18***b* and extend in a zigzag line shape.
(91) FIG. **21** is an enlarged plan view of a coupling wire in a wiring region according to a tenth modification. As illustrated in FIG. **21**, a coupling wire **17***b* according to the tenth modification has a meander shape. More specifically, the coupling wire **17***b* has a plurality of arc portions **18***c* and a plurality of linear portions **18***d*, and the arc portions **18***c* and the linear portions **18***d* are alternately coupled and extend in a meander shape.
(92) FIG. **22** is an enlarged plan view of a coupling wire in a wiring region according to an eleventh modification. As illustrated in FIG. **22**, a coupling wire **17***c* according to the eleventh modification has a meander shape. More specifically, the coupling wire **17***c* has a plurality of arc portions **18***e* and **18***f* and a plurality of linear portions **18***g*. The arc portions **18***e* and **18***f* are connected in an S-shape, and the connected arc portions **18***e* and **18***f* are coupled by the linear portions **18***g* and extend in a meander shape.
(93) While FIG. **20** (ninth modification) to FIG. **22** (eleventh modification) illustrate the single coupling wires **17***a*, **17***b*, and **17***c*, respectively, a plurality of coupling wires **17***a*, **17***b*, and **17***c* are arrayed in the wiring region WA as illustrated in FIG. **9** and other figures. The coupling wires **17***a*, **17***b*, and **17***c* described in the ninth to the eleventh modifications are applicable to the first embodiment and the first to the eighth modifications described above.
(94) The linear coupling wire **17** and at least one of the coupling wires **17***a*, **17***b*, and **17***c* may be combined in one wiring region WA. For example, the linear coupling wire **17** may be provided at the center of the wiring region WA in the first direction Dx where the stress is relatively small, and the zigzag-line-shaped coupling wire **17***a* or the meander-shaped coupling wire **17***b* or **17***c* may be provided at the outer ends of the wiring region WA in the first direction Dx where the stress is relatively large. With this configuration, the detection device **1** can effectively suppress the stress acting on the coupling wire **17**.
(95) FIG. **23** is an enlarged plan view of a wiring region according to a twelfth modification. As illustrated in FIG. **23**, the distance between the coupling wires **17** adjacently disposed in the first direction Dx according to the twelfth modification is larger at the outer ends of the wiring region WA of the substrate **21** in the first direction Dx than at the center in the first direction Dx. In FIG. **23**, the distance between the coupling wires **17** gradually increases from the center of the wiring region WA of the substrate **21** in the first direction Dx to the outer ends in the first direction Dx. The number of through holes TH formed between the coupling wires **17** adjacently disposed in the first direction Dx is larger at the outer ends of the wiring region WA of the substrate **21** in the first direction Dx than at the center of the wiring region WA in the first direction Dx.
(96) In other words, the arrangement density of the coupling wires **17** is smaller at the outer ends of the wiring region WA of the substrate **21** in the first direction Dx than at the center of the wiring region WA in the first direction Dx. The arrangement density of the through holes TH is larger at

the outer ends of the wiring region WA of the substrate **21** in the first direction Dx than at the center of the wiring region WA in the first direction Dx.

(97) With this configuration, the twelfth modification can effectively relieve the stress generated in the substrate **21** and suppress the stress acting on the coupling wires **17** at the outer ends of the wiring region WA in the first direction Dx. The configuration illustrated in FIG. **23** can be combined with the first to the eleventh modifications described above.

(98) FIG. **24** is sectional view of a force sensor according to a thirteenth modification including parallel electrodes. FIGS. **24** to **29** do not illustrate the circuit layer **55** (the drive transistor Tr and the insulating layer **23**). As illustrated in FIG. **24**, a force sensor **3**A according to the thirteenth modification includes parallel electrodes in which a detection electrode **31**A and a common electrode **32**A are disposed on the same plane. With the parallel electrodes (the detection electrode **31**A and the common electrode **32**A), an electric current flows in the planar direction in a sensor layer **30**A (refer to the arrows in FIG. **24**). The detection electrode **31**A and the common electrode **32**A can have any planar shape. The detection electrode **31**A and the common electrode **32**A can have a comb shape, for example.

(99) FIG. **25** is a sectional view of a force sensor according to a fourteenth modification including hybrid electrodes. As illustrated in FIG. **25**, a force sensor **3**B according to the fourteenth modification includes hybrid electrodes obtained by combining counter electrodes and parallel electrodes. In the hybrid electrodes, a detection electrode **31**B and a common electrode **32**B are disposed on the upper surface of the substrate **21**. In addition, an intermediate electrode **34** is disposed on the lower surface of the protective layer **33**. The intermediate electrode **34** faces the detection electrode **31**B and the common electrode **32**B.

(100) When force is input to the force sensor **3**B including the hybrid electrodes, an electric current first flows from the common electrode **32**B to the intermediate electrode **34**. The electric current then flows in the planar direction along the intermediate electrode **34**. Subsequently, the electric current flows to the detection electrode **31**B via a sensor layer **30**B. The detection electrode **31**B, the common electrode **32**B, and the intermediate electrode **34** can have any planar shape. The detection electrode **31**A and the common electrode **32**A can have a comb shape, for example. The intermediate electrode **34** may be a solid film extending over the detection region AA.

(101) FIG. **26** is a sectional view of a force sensor according to a fifteenth modification before force is input. FIG. **27** is a sectional view of the force sensor according to the fifteenth modification after force is input. As illustrated in FIG. **26**, a sensor layer **30**C of a force sensor **3**C according to the fifteenth modification is provided on the lower surface (surface facing a detection electrode **31**C) of a common electrode **32**C. The sensor layer **30**C is made of conductive resin. Unlike the sensor layers **30**, **30**A, and **30**B described above, the sensor layer **30**C does not have the protrusions ***30**a*, and the surface facing the detection electrode **31**C is flat. The sensor layer **30**C faces the detection electrode **31**C with a space interposed therebetween and is not in contact with the detection electrode **31**C before force is input.

(102) As illustrated in FIG. **27**, when the force is input, the sensor layer **30**C is deformed to be recessed in the force direction A**1** and comes into contact with the detection electrode **31**C. As a result, an electric current flows to the detection electrode **31**C. As the force increases, and the contact area between the sensor layer **30**C and the detection electrode **31**C increases, the electric current flowing to the detection electrode **31**C increases.

(103) FIG. **28** is a sectional view of a force sensor according to a sixteenth modification before force is input. FIG. **29** is a sectional view of the force sensor according to the sixteenth modification after force is input. As illustrated in FIG. **28**, a sensor layer **30**D of a force sensor **3**D according to the sixteenth modification includes a resin layer **35** and conductive fillers **36** dispersed in the resin layer **35**. The resin layer **35** of the sensor layer **30**D is provided to fill the space between a detection electrode **31**D and a common electrode **32**D facing each other and is in contact with the detection electrode **31**D and the common electrode **32**D.

(104) Most of the conductive fillers **36** of the sensor layer **30**D are not in contact with the detection electrode **31**D or the common electrode **32**D before force is input. While some of the conductive fillers **36** are in contact with the detection electrode **31**D and the common electrode **32**D, no conductive path is formed between the detection electrode **31**D and the common electrode **32**D, and the detection electrode **31**D and the common electrode **32**D facing across the sensor layer **30**D remain insulated.

(105) As illustrated in FIG. **29**, when the force is input, the protective layer **33** and the common electrode **32**D move in the force direction A**1**, and the conductive fillers **36** of the sensor layer **30**D come into contact with the detection electrode **31**D and the common electrode **32**D. The conductive fillers **36** themselves come into contact with each other. As a result, a conductive path is formed between the detection electrode **31**D and the common electrode **32**D, and an electric current flows to the detection electrode **31**D. As the force increases, and the contact area between the conductive fillers **36** of the sensor layer **30**D and the detection electrode **31**D and the common electrode **32**D increases, the electric current flowing to the detection electrode **31**D increases.

(106) FIG. **30** is a sectional view of a detection device according to a seventeenth modification. As illustrated in FIG. **30**, a detection device **1**A according to the seventeenth modification includes resin layers **24** and **25** provided in the wiring region WA. The resin layers **24** and **25** are made of elastic resin material. The resin layer **24** is provided on the upper surface of the insulating layer **23** in the wiring region WA. The resin layer **25** is provided on the lower surface of the substrate **21** in the wiring region WA. In the force direction A**1**, the substrate **21**, the coupling wire **17**, and the insulating layer **23** are provided between the resin layer **24** and the resin layer **25**.

(107) The through holes TH, which are not illustrated in FIG. **30**, are formed passing through the resin layers **24** and **25**. With the elastic resin layers **24** and **25**, the present modification can protect the coupling wire **17** while suppressing the stress generated in the wiring region WA. Both the resin layers **24** and **25** are not necessarily provided, and only one of them may be provided. While the resin layers **24** and **25** are provided only in the wiring region WA in FIG. **30**, they may also be provided in part of the peripheral region GA and the coupling region CA.

(108) FIG. **31** is a plan view of a detection device according to an eighteenth modification. As illustrated in FIG. **31**, a detection device **1**B according to the eighteenth modification has two wiring regions WAa and WAb and two coupling regions CAa and CAb. The two wiring regions WAa and WAb are provided to one side of the peripheral region GA and are disposed side by side in the first direction Dx with a gap interposed therebetween. The coupling regions CAa and CAb are coupled to the wiring regions WAa and WAb, respectively. The wiring region WAa according to the present modification is positioned between the coupling region CAa, and the detection region AA and the peripheral region GA in the second direction Dy. The wiring region WAb is positioned between the coupling region CAb, and the detection region AA and the peripheral region GA in the second direction Dy. The detection region AA, the peripheral region GA, the two wiring regions WAa and WAb, and the two coupling regions CAa and CAb are formed on the common substrate **21**.

(109) The coupling region CAa of the substrate **21** is coupled to a wiring substrate **110***a* via a plurality of terminals Ta. The coupling region CAb of the substrate **21** is coupled to a wiring substrate **110***b* via a plurality of terminals Tb. With this configuration, the detection device **1**B according to the present modification can increase the flexibility in coupling with the external wiring substrates **110***a* and **110***b*. While the wiring substrates **110***a* and **110***b* are coupled to the coupling regions CAa and CAb, respectively, in this example, the present modification is not limited thereto. Alternatively, the drive IC **102** (refer to FIG. **1**) may be mounted on at least one of the coupling regions CAa and CAb.

(110) FIG. **32** is a plan view of a detection device according to a nineteenth modification. As illustrated in FIG. **32**, a detection device **1**C according to the nineteenth modification has two wiring regions WAc and WAd and two coupling regions CAc and CAd. In FIG. **32**, the wiring

regions WAc and WAd are hatched to make the drawing easier to see.

(111) The detection region AA and the peripheral region GA, the wiring region WAc, the coupling region CAc, the wiring region WAd, and the coupling region CAd are positioned in this order in the second direction Dy. In other words, the wiring region WAc is positioned between the coupling region CAc, and the detection region AA and the peripheral region GA in the second direction Dy. The wiring region WAc is also provided to the parts adjacent to the coupling region CAc in the first direction Dx. The drive IC **102** is mounted on the coupling region CAc.

(112) The wiring region WAd is positioned between the coupling region CAc and the coupling region CAd in the second direction Dy. The wiring region WAd is also provided to the parts adjacent to the coupling region CAd in the first direction Dx. The coupling region CAd is coupled to the wiring substrate **110** via the terminals T. The width in the first direction Dx decreases in the order of the wiring region WAc, the coupling region CAc, the wiring region WAd, and the coupling region CAd.

(113) FIG. **33** is a plan view of an example of the use of the detection device according to the nineteenth modification. As illustrated in FIG. **33**, the detection device **1**C according to the nineteenth modification is attached to a cylindrical housing **120**A having a hollow shape. The detection region AA and the peripheral region GA of the substrate **21** are attached in a manner curved along the outer peripheral surface of the housing **120**A.

(114) The wiring region WAc of the substrate **21** is folded back at the opening end of the housing **120**A. In other words, one end of the wiring region WAc is coupled to the peripheral region GA on the outer peripheral surface of the housing **120**A, and the other end of the wiring region WAc is coupled to the coupling region CAc on the inner peripheral surface of the housing **120**A. The coupling region CAc, the wiring region WAd, and the coupling region CAd of the substrate **21** are disposed inside the housing **120**A. In other words, the drive IC **102** mounted on the coupling region CAc and the wiring substrate **110** coupled to the coupling region CAd are also disposed inside the housing **120**A.

(115) FIG. **34** is a sectional view along line XXXIV-XXXIV′ of FIG. **33**. As illustrated in FIG. **34**, a base **121**A is provided on the inner peripheral surface of the housing **120**A. The coupling region CAd of the substrate **21** is attached on the flat surface of the base **121**A. With this configuration, the coupling region CAd is provided on the inner peripheral surface of the housing **120** in a flat state, thereby securing satisfactory coupling with the wiring substrate **110**.

Second Embodiment

(116) FIG. **35** is a sectional view of the detection device according to a second embodiment. While the first embodiment and the modifications described above include the force sensors **3** as the detection elements, for example, the detection elements are not limited to the force sensors **3**. As illustrated in FIG. **35**, a detection device **1**D according to the second embodiment includes optical sensors **4** (detection elements) provided in the detection region AA to detect light incident on them. In the detection device **1**D according to the second embodiment, the peripheral region GA, the wiring region WA, and the coupling region CA can employ the configurations described in the first embodiment and the modifications.

(117) The optical sensor **4** is provided on the insulating layer **23** of the circuit layer **55**. The optical sensor **4** includes a sensor layer **40**, a detection electrode **41**, and a common electrode **42**. At least one of the detection electrode **41** and the common electrode **42** is made of translucent conductive material, such as indium tin oxide (ITO). The sensor layer **40** is made of photoelectric conversion material. The sensor layer **40** is an organic optical sensor composed of an electron transport layer, a light emission layer, and a hole transport layer, for example. Alternatively, the sensor layer **40** may be a positive intrinsic negative (PIN) photodiode formed by a semiconductor layer, such as silicon.

(118) A protective layer **43** is provided covering the common electrode **42**. Light from the outside is transmitted from the upper surface of the protective layer **43**, that is, the input surface **1***a* to the sensor layer **40** of the optical sensor **4**.

(119) A plurality of optical sensors **4** are arrayed in a matrix (row-column configuration) in the detection region AA in plan view, which is not illustrated. The detection electrodes **41** are arrayed in a matrix (row-column configuration) in the detection region AA in plan view and are separated from each other corresponding to the respective optical sensors **4**. The common electrode **42** is provided over the entire surface of the detection region AA to cover the optical sensors **4**.
(120) Similarly to the first embodiment and the modifications, the detection device **1**D according to the present embodiment is used by being attached to the solid housing **120** or the hollow cylindrical housing **120**A. The detection device **1**D can be used for pulse measurement and vein authentication in a gripping state where an object to be detected grips the housing **120** or **120**A.
(121) While exemplary embodiments according to the present invention have been described, the embodiments are not intended to limit the invention. The contents disclosed in the embodiments are given by way of example only, and various modifications may be made without departing from the spirit of the present invention. Appropriate modifications made without departing from the spirit of the present invention naturally fall within the technical scope of the invention. At least one of various omissions, substitutions, and modifications of the components may be made without departing from the gist of the embodiments and the modifications described above.

Claims

1. A detection device comprising: a substrate having a detection region, a coupling region, and a wiring region between the detection region and the coupling region; a plurality of sensors disposed in the detection region of the substrate; a plurality of terminals disposed in the coupling region of the substrate and electrically coupled to the sensors; a plurality of coupling wires arranged with a gap interposed therebetween in the wiring region of the substrate and electrically coupling the sensors to the terminals; an insulating layer provided to the wiring region of the substrate and covering the coupling wires; and a plurality of through holes formed in the wiring region of the substrate, the through holes passing through from an upper surface of the insulating layer to a lower surface of the substrate, in a thickness direction between the coupling wires.
2. The detection device according to claim 1, further comprising: a housing having a curved surface; and a base provided to the housing and having a flat surface, wherein the detection region and the wiring region of the substrate are provided to the curved surface of the housing, and the coupling region is provided to the flat surface of the base.
3. The detection device according to claim 1, wherein the through holes are arrayed along an extending direction of the coupling wires.
4. The detection device according to claim 1, wherein the through holes have a larger opening area as the distance between the coupling wires adjacently disposed is larger.
5. The detection device according to claim 1, wherein a larger number of the through holes are formed as the distance between the coupling wires adjacently disposed is larger.
6. The detection device according to claim 1, wherein the through holes have a circular, elliptical, or rectangular shape.
7. The detection device according to claim 1, wherein the through holes have a slit shape.
8. The detection device according to claim 1, wherein the through holes have a slit shape, and the through holes each comprise: a first slit portion extending along an extending direction of the coupling wires; and a second slit portion intersecting the first slit portion.
9. The detection device according to claim 1, wherein the coupling wires are adjacently disposed in a first direction with a gap interposed therebetween, and an opening width of the through holes in the first direction is substantially equal to the distance between the coupling wires adjacently disposed in the first direction.
10. The detection device according to claim 1, wherein the coupling wires have a meander shape or a zigzag line shape.

11. The detection device according to claim 1, wherein the coupling wires are arranged in a first direction with a gap interposed therebetween, and the distance between the coupling wires adjacently disposed in the first direction is larger at an outer end of the wiring region of the substrate in the first direction than at a center of the wiring region in the first direction.
12. The detection device according to claim 1, wherein the coupling wires are adjacently disposed in a first direction with a gap interposed therebetween, and the number of the through holes formed between the coupling wires adjacently disposed in the first direction is larger at an outer end of the wiring region of the substrate in the first direction than at a center of the wiring region in the first direction.
13. The detection device according to claim 1, wherein the sensors are force sensors configured to detect force applied to the force sensors.
14. The detection device according to claim 1, wherein the sensors are optical sensors configured to detect light incident on the optical sensors.